

Beta and gamma function

$$\int_0^{\infty} e^{-x} x^{n-1} dx = \Gamma(n)$$

$$\int_0^{\infty} e^{-x} x^n dx = \Gamma(n+1)$$

* Basically we defined $\Gamma(n)$ for extending factorials for $x \in \mathbb{R}$

$$\Gamma(z+1) = z! \quad \Gamma(n) = (n-1)!$$

① + property $\Gamma(n) = \frac{\Gamma(n+1)}{n}$

② - property $\Gamma(n) = \Gamma(n-1)(n-1)$

$$\Gamma(0) = \infty$$

$$\Gamma(\text{any negative int}) = \infty$$

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

• Problem Solving

① Type 1 : Integrals of form $\int_0^{\infty} e^{-ax^n} dx = \frac{1}{n(a)^{1/n}} \Gamma\left(\frac{1}{n}\right)$

Que 1) $\int_0^{\infty} e^{-x^4} dx$

$e^{-x^4} dx$, let hamisha $x^n = t$ aur $dx = t^{1/n}$

$$x^4 = t$$

$$x = t^{1/4}$$

when $x=0 \quad t=0$
 $x=\infty \quad t=\infty$

$$\frac{dx}{dt} = \frac{1}{4} t^{-3/4} \frac{dt}{dx}$$

$$dx = \frac{1}{4} t^{-3/4} dt$$

$$\int_0^{\infty} e^{-t} \frac{1}{4} t^{-3/4} dt = \frac{1}{4} \int_0^{\infty} e^{-t} t^{-3/4} dt$$

$$\int_0^{\infty} e^{-x} x^{n-1} dx = \Gamma(n)$$

$$n-1 = -\frac{3}{4}$$

$$= \frac{1}{4} \Gamma\left(\frac{1}{4}\right)$$

② Type 2 : $\int_0^{\infty} e^{-ax^n} x^m dx = \frac{1}{n} \frac{1}{a^{m+1/n}} \Gamma\left(\frac{m+1}{n}\right)$

again substitute $x^n = t$

que) Evaluate $\int_0^{\infty} x e^{-x^4} dx$

$$x^4 = t$$

$$x = t^{1/4}$$

$$dx = \frac{1}{4} t^{-3/4} dt$$

$$\Rightarrow \frac{1}{4} \int t^{1/4} t^{-3/4} dt e^{-t} \Rightarrow \frac{1}{4} \int_0^{\infty} e^{-t} t^{-1/2} dt$$

$$\Rightarrow \frac{1}{4} \int_0^{\infty} e^{-t} t^{(-1/2+1)-1} dt \Rightarrow \frac{1}{4} \Gamma\left(\frac{1}{2}\right) = \frac{\sqrt{\pi}}{4}$$

Que 2) $\left(\int_0^\infty x \cdot e^{-x^8} dx \right) \cdot \left(\int_0^\infty x^3 \cdot e^{-x^4} dx \right)$

$$x^8 = t$$

$$x = t^{1/8}$$

$$dx = \frac{1}{8} t^{-7/8} dt$$

$$x^4 = t$$

$$x = t^{1/4}$$

$$dx = \frac{1}{4} t^{-3/4} dt$$

$$\frac{2 \times 1}{2 \times 2} = \frac{2}{4}$$

$$\frac{2-2}{4}$$

$$\left(\int_0^\infty t^{1/8} \cdot e^{-t} \cdot \frac{1}{8} t^{-7/8} dt \right) \cdot \left(\int_0^\infty t^{1/2} \cdot e^{-t} \cdot \frac{1}{4} t^{-3/4} dt \right)$$

$$\frac{1}{32} \cdot \left(\int_0^\infty e^{-t} \cdot t^{-3/4} dt \right) \cdot \left(\int_0^\infty e^{-t} \cdot t^{-1/4} dt \right)$$

$$\left[\Gamma\left(\frac{1}{4}\right) \cdot \Gamma\left(\frac{3}{4}\right) \right] \frac{1}{32}$$

$$\Gamma\left(\frac{1}{4}\right) \Gamma\left(\frac{3}{4}\right) = \sqrt{2}\pi$$

$$\frac{\sqrt{2}\pi}{\sqrt{2} \times 16 \sqrt{2}} = \frac{\pi}{16\sqrt{2}}$$

③ Type 3 : $\int_0^1 x^m (\log x)^n dx = \frac{(-1)^n}{(m+1)^{n+1}} \Gamma(n+1)$ $m=n$ bhi ho sakta hai

Simple trick $\log x = -t$, put $x = e^{-t}$ $\log x = -t$

que) $\int_0^1 (x \log x)^4 dx = \int_0^1 x^4 (\log x)^4 dx$

$$\log x = -t$$

$$x = e^{-t}$$

$$x=0 \quad t=\infty$$

$$x=1 \quad t=0$$

$$dx = -e^{-t} dt$$

$$+ \int_0^\infty e^{-4t} t^4 e^{-t} dt$$

∞ se 0?

- लगाई!

$$= - \int_0^\infty e^{-5t} t^4 dt$$

$$\text{now } 5t = z$$

$$= - \int_0^\infty e^{-z} \cdot \left(\frac{z}{5}\right)^4 \frac{dz}{5}$$

$$= - \frac{1}{5^5} \int_0^\infty e^{-z} \cdot z^4 dz = + \frac{1}{5^5} \Gamma(5) = \frac{4!}{5^5} = \frac{24}{5^5}$$

$$\text{ex 4: } \int_0^{\infty} \frac{x^a}{a^x} dx = \frac{1}{(\log a)^{a+1}} \Gamma(a+1)$$

$$\text{put } a^x = e^t$$

$$\int_0^{\infty} \frac{x^7}{7^x} dx$$

$$7^x = e^t$$

$$x \ln 7 = t$$

$$x = \frac{t}{\ln 7}$$

$$\frac{dx}{dt} = \frac{1}{\ln 7} \Rightarrow dx = \frac{dt}{\ln 7}$$

$$\int_0^{\infty} \frac{t^7}{(\ln 7)^7} e^{-t} \frac{dt}{\ln 7}$$

$$\frac{1}{(\ln 7)^8} \int_0^{\infty} e^{-t} t^7 dt$$

$$\frac{1}{(\ln 7)^8} \Gamma(8) = \frac{7!}{(\ln 7)^8}$$

Random ques

Que 1) Evaluate $\int_0^{\infty} (x)^{1/4} \cdot e^{-x^{1/2}} dx$

$$x^{1/2} = t$$

$$x = t^2$$

$$\frac{dx}{dt} = 2t$$

$$dx = 2t dt$$

$$= \int_0^{\infty} t^{1/2} \cdot e^{-t} 2t dt$$

$$= 2 \int_0^{\infty} t^{3/2} e^{-t} dt = 2 \Gamma\left(\frac{5}{2}\right)$$

$$= 2 \times \frac{5}{2} \times \frac{3}{2} \Gamma\left(\frac{1}{2}\right)$$

$$= \frac{15}{2} \sqrt{\pi}$$

Que 2) $\int_0^{\infty} \frac{dx}{e^{3+4x^2}}$

$$3+4x^2 = e^t$$

$$4x^2 \ln 3 = t$$

$$x^2 = \frac{t}{4 \ln 3}$$

$$2x = \frac{dt}{dx} \left(\frac{1}{4 \ln 3} \right)$$

$$x dx = \frac{dt}{8 \ln 3}$$

Que 3) $\int_0^{\infty} x^a \cdot e^{-2x^2} dx$

$$2x^2 = t$$

$$x = \sqrt{\frac{t}{2}}$$

$$\frac{dx}{dz} = \frac{1}{2\sqrt{t}} \cdot \frac{1}{\sqrt{2}} \frac{dt}{dz}$$

$$dx = \frac{dt}{2\sqrt{2}\sqrt{t}}$$

$$\int_0^{\infty} \left(\frac{t}{2}\right)^{a/2} e^{-t} \frac{dt}{2\sqrt{2}} t^{1/2}$$

$$\frac{1}{2^{a/2} \cdot 2 \cdot 2^{1/2}} \int_0^{\infty} (t^{a/2-1/2}) \cdot e^{-t} dt$$

$$\frac{1}{2^{\frac{a+3}{2}}} \Gamma\left(\frac{a+1}{2}\right)$$

for $a=9$

$$\frac{1}{2^6} \Gamma(5) = \frac{24^6}{4 \times 4 \times 4} = \frac{6}{16}$$



function

'We are removing, What is giving us Rain'

$$\beta(m, n) = \int_0^1 x^m (1-x)^n dx$$

$$\beta(m+1, n+1) = \int_0^1 x^m (1-x)^n dx$$

Properties of beta function

1) $\beta(m, n) = \beta(n, m)$

2) Relationship between beta and gamma fcn

$$\beta(m, n) = \frac{\Gamma(m) \cdot \Gamma(n)}{\Gamma(m+n)} = \frac{(m-1)! (n-1)!}{(m+n-1)!}$$

3) $\Gamma(p) \Gamma(1-p) = \frac{\pi}{\sin p\pi}$

4) Duplication formula

$$\Gamma(m) \Gamma(m + \frac{1}{2}) = \frac{\sqrt{\pi} \cdot \Gamma(2m)}{2^{2m-1}}$$

ex) $\Gamma(\frac{1}{4}) \Gamma(\frac{3}{4})$ $\Gamma(\frac{1}{4}) \Gamma(\frac{1}{4} + \frac{1}{2}) = \frac{\sqrt{\pi} \Gamma(2 \times \frac{1}{4})}{2^{2 \times \frac{1}{4} - 1}} = \frac{\sqrt{\pi} \times \sqrt{\pi}}{2^{-1/2}} = \pi \sqrt{2}$

• Problem Solving

① Type 1 : $\int_0^a x^m (a-x)^n dx = a^{m+n+1} \beta(m+1, n+1)$

करना क्या है : put $x = at$

que 1) $\int_0^4 \sqrt{x} (4-x)^{3/2} dx$

$x = 4t$

$\frac{dx}{dt} = 4$

$dx = 4 dt$

$x=0 \quad t=0$
 $x=4 \quad t=1$

$$4 \int_0^1 2 t^{1/2} (4-4t)^{3/2} dt$$

$$\Rightarrow 64 \int_0^1 t^{1/2} (1-t)^{3/2} dt$$

$$\Rightarrow 64 \beta(\frac{3}{2}, \frac{5}{2}) = \frac{64 (\frac{3}{2}) \sqrt{\pi} \times \frac{5 \times 3}{2} \times \sqrt{\pi}}{6}$$

② Type 2 : $\int_0^1 x^m (1-x^n)^p dx = \frac{1}{n} \beta(\frac{m+1}{n}, p+1)$

करना क्या है : put $x^n = t$ and $x = t^{1/n}$

que 1) $\int_0^1 x^m (1-x^n)^p dx \rightarrow \int_0^1 x^6 (1-x^2)^{1/2} dx$

$x^2 = t$

$2x dx = dt$

$x = \sqrt{t}$

$\frac{dx}{dt} = \frac{1}{2\sqrt{t}} \Rightarrow dx = \frac{dt}{2\sqrt{t}}$

$$\frac{1}{2} \int_0^1 t^3 (1-t)^{1/2} \frac{dt}{t^{1/2}} = \int_0^1 t^{(2 \times \frac{1}{2})+1} (1-t)^{1/2} dt$$

$$= \frac{1}{2} \beta(\frac{7}{2}, \frac{3}{2})$$

Que) $\left(\int_0^3 \frac{x^{3/2}}{\sqrt{3-x}} dx \right) \cdot \left(\int_0^1 \frac{dx}{\sqrt{1-x^{1/4}}} \right) = \frac{432}{35} \pi$

$\left(\int_0^3 x^{3/2} (3-x)^{-1/2} dx \right) \cdot \left(\int_0^1 x^0 (1-x^{1/4})^{-1/2} dx \right)$

$x = 3t$

$\frac{dx}{dt} = 3$

$dx = 3 dt$

$x^{1/4} = t$

$x = t^4$

$dx = 4t^3 dt$

$\left(3 \cdot 3^{3/2} \cdot 3^{-1/2} \int_0^1 t^{3/2} (1-t)^{-1/2} dt \right) \left(4 \int_0^1 t^3 (1-t)^{-1/2} dt \right)$

$36 \cdot \beta\left(\frac{5}{2}, \frac{1}{2}\right) \cdot \beta\left(4, \frac{1}{2}\right) = 36 \cdot \frac{\frac{3}{2} \sqrt{\pi} \sqrt{\pi}}{2} \cdot \frac{3 \times 2 \times \sqrt{\pi}}{\frac{7}{2} \times \frac{5}{2} \times \frac{3}{2} \times \sqrt{\pi}}$

$= \left(\frac{36 \times 3 \times 3 \times 2 \times 2 \times 2 \times 2}{7 \times 5 \times 3 \times 2} \right) \pi$

$= \left(\frac{5184}{210} \right) \pi$

③ Type 3 : $\int_0^1 (1-x^{1/n})^m dx = \frac{m! n!}{(m+n)!}$

Put $x^{1/n} = t \Rightarrow x = t^n$

Que) $\int_0^1 (1-x^{1/3})^{11/2} dx$

$x^{1/3} = t \Rightarrow x = t^3$

$dx = 3t^2 dt$

$3 \int_0^1 t^2 (1-t)^{11/2} dt = 3 \beta\left(3, \frac{13}{2}\right)$

④ Type 4 : $\int_0^{\pi/2} \sin^p \theta \cdot \cos^q \theta d\theta = \frac{1}{2} \beta\left(\frac{p+1}{2}, \frac{q+1}{2}\right)$

⑤ Type 5: $\int_0^{\infty} \frac{x^m}{(1+x^2)^n} dx = \frac{1}{2} B\left(\frac{m+1}{2}, \frac{2n-m-1}{2}\right)$

* क्या करना है : put $x = \tan \theta$ *

Que) $\int_0^{\infty} \frac{dx}{(1+x^2)^{7/2}}$

$x = \tan \theta$

$\frac{dx}{d\theta} = \sec^2 \theta$

$dx = \sec^2 \theta d\theta$

$x=0 \quad \theta=0$
 $x=\infty \quad \theta=\pi/2$

$\int_0^{\pi/2} \frac{\sec^2 \theta d\theta}{(\sec^2 \theta)^{7/2}} =$

$\frac{\pi}{2} \int_0^{\pi/2} (\sec^2 \theta)^{1-7/2} d\theta$

$= \frac{\pi}{2} \int_0^{\pi/2} (\sec^2 \theta)^{-5/2} d\theta$

$\sec \theta = z$

$\frac{d \sec \theta}{d\theta} = dz$

$= \frac{\pi}{2} \int_0^{\pi/2} \frac{1}{(\sec \theta)^5} d\theta$

$= \frac{\pi}{2} \int_0^{\pi/2} (\cos \theta)^5 d\theta$

⑥ Type 6 : $\int_0^{\infty} \frac{x^m}{(1+x)^n} = \beta(m+1, n-m-1)$

* Jabhi bhi किसी सवाल में $(1+x)^n$ आये, तब ये लगाना।

Formula लगाओ direct $\int_0^{\infty} \frac{x^m}{(1+x)^n} = \beta(m+1, n-m-1)$

⑦ Type 7 : $\int_a^b (x-a)^m (b-x)^n dx = (b-a)^{m+n+1} \beta(m+1, n+1)$ $\frac{1}{m} \sqrt{(x-a)(b-x)}$ $\leftarrow$ m can be equal to n

Formula लगाओ Answer पाओ।

$(x - (\text{lower limit})) ((\text{upper limit}) - x)$

⑧ Type 8 :

$$\int (I)(II) = I \int II - \int \left(\frac{dI}{dx} \right) \int II dx$$

$$\int (uv) dx = u \int v dx - \int \left(\frac{du}{dx} \right) \int v dx dx \quad \xrightarrow{\text{ILATE}} 1^{st}$$

$$\begin{aligned} \int x \tan^{-1} x &= (\tan^{-1} x) \frac{x^2}{2} - \int \left(\frac{1}{1+x^2} \right) \frac{x^2}{2} dx \\ &= (\tan^{-1} x) \frac{x^2}{2} - \frac{1}{2} \int \left(\frac{x^2+1}{1+x^2} - \frac{1}{1+x^2} \right) dx \\ &= (\tan^{-1} x) \frac{x^2}{2} - \frac{1}{2} (x - \tan^{-1}(x)) \\ &= (\tan^{-1} x) \frac{x^2}{2} - \frac{x}{2} + \frac{\tan^{-1} x}{2} \end{aligned}$$

Shortcut

$x e^x$

D	I
x	e^x
1	e^x
0	e^x

$x e^x - e^x$

D(I)	I(II)
$+ x^2$	$\cos 2x$
$- 2x$	$\frac{1}{2} \sin 2x$
$+ 2$	$-\frac{1}{4} \cos 2x$
$- 0$	$-\frac{1}{8} \sin 2x$

$x^2 \cos 2x$
(I) (II)

$\frac{x^2 \sin 2x}{2} + \frac{1}{2} x \cos 2x - \frac{1}{4} \sin 2x$

$x^3 e^{2x}$

$+ x^3$	e^{2x}
$- 3x^2$	$\frac{1}{2} e^{2x}$
$+ 6x$	$\frac{1}{4} e^{2x}$
$- 6$	$\frac{1}{8} e^{2x}$
0	$\frac{1}{16} e^{2x}$

$\frac{x^3 e^{2x}}{2} - \frac{3}{4} x^2 e^{2x} + \frac{6}{8} x e^{2x} - \frac{6}{16} e^{2x}$

